

A Survey on RRAM Technology and MOS Integration

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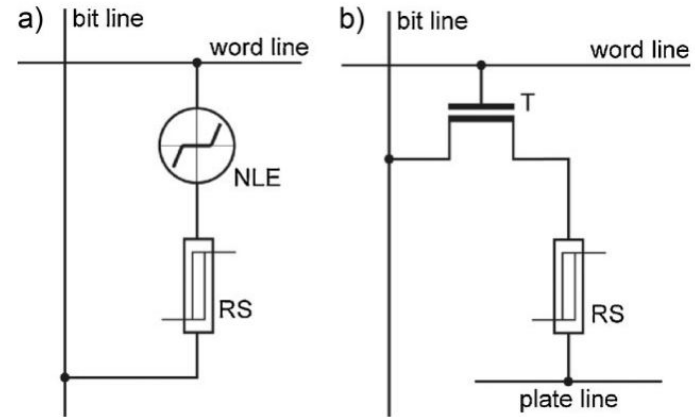
Overview

- What is RRAM and what are VCMs in particular?
- Why and where would you use RRAM?
- How are these devices actually used in CMOS?
- Technology dependent challenges (Planar -> FD-SOI -> FinFET)

What is RRAM?

RRAM - Resistive Random Access Memory

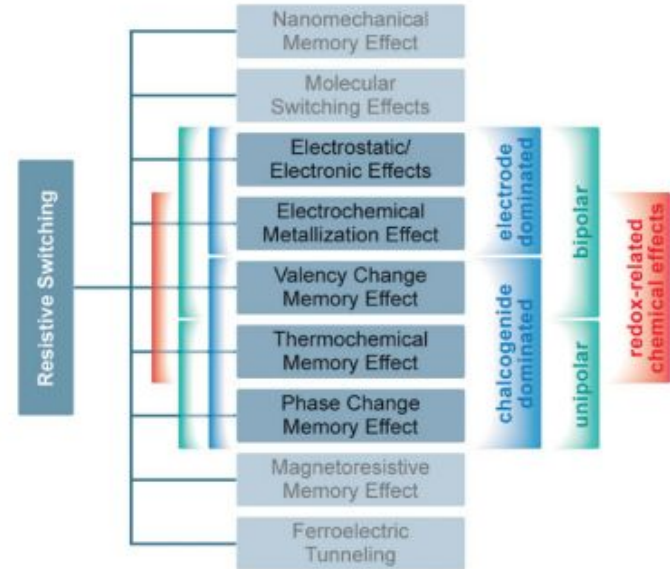
- A form of Non-volatile memory (NVM).
- Uses a resistive switching device to store information.
- Resistance changes based on voltage applied.
- Stores a logic 0 in the high resistance state (HRS) and a logic 1 in the low resistance state (LRS).



Waser, R., Dittmann, R., Staikov, G. and Szot, K. (2009), Redox-Based Resistive Switching Memories – Nanoionic Mechanisms, Prospects, and Challenges. Adv. Mater., 21: 2632-2663. <https://doi.org/10.1002/adma.200900375>

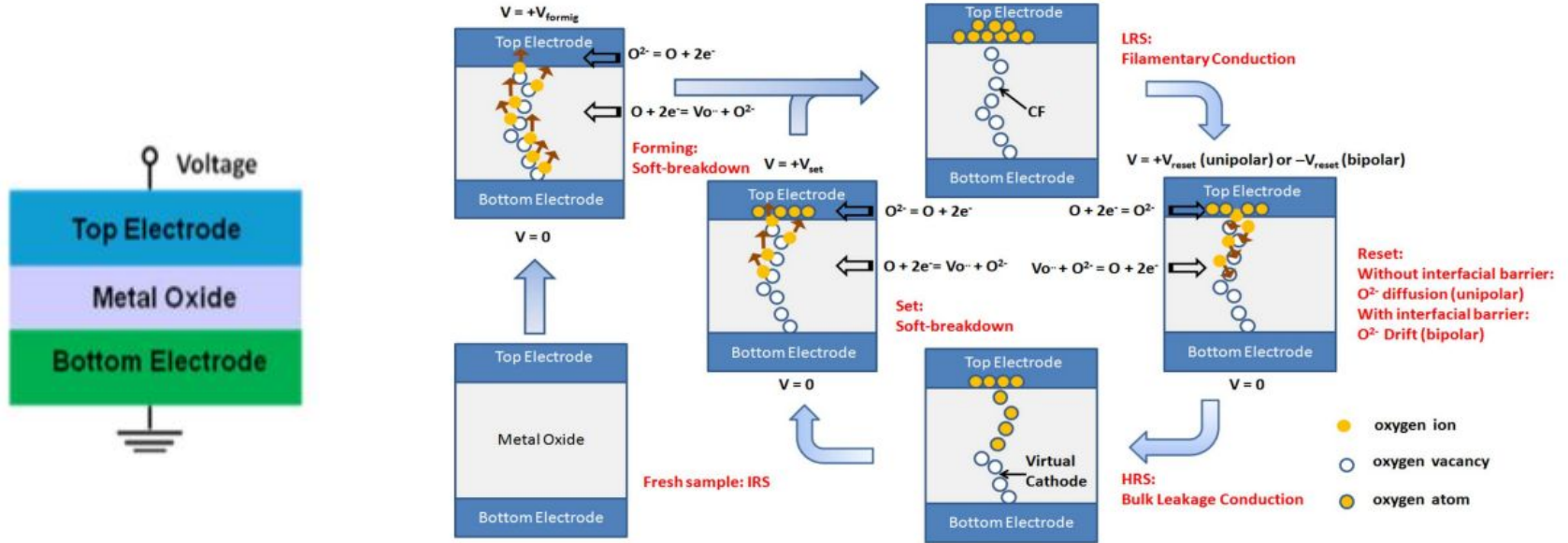
Resistive Switching Devices

- Industry has converged to Valence Change Memory's (VCMs).
- HfO_2 and TaO_x being the front runners.



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How do VCMs Work?



H. . -S. P. Wong et al., "Metal-Oxide RRAM," in Proceedings of the IEEE, vol. 100, no. 6, pp. 1951-1970, June 2012, doi: 10.1109/JPROC.2012.2190369.

Why RRAM?

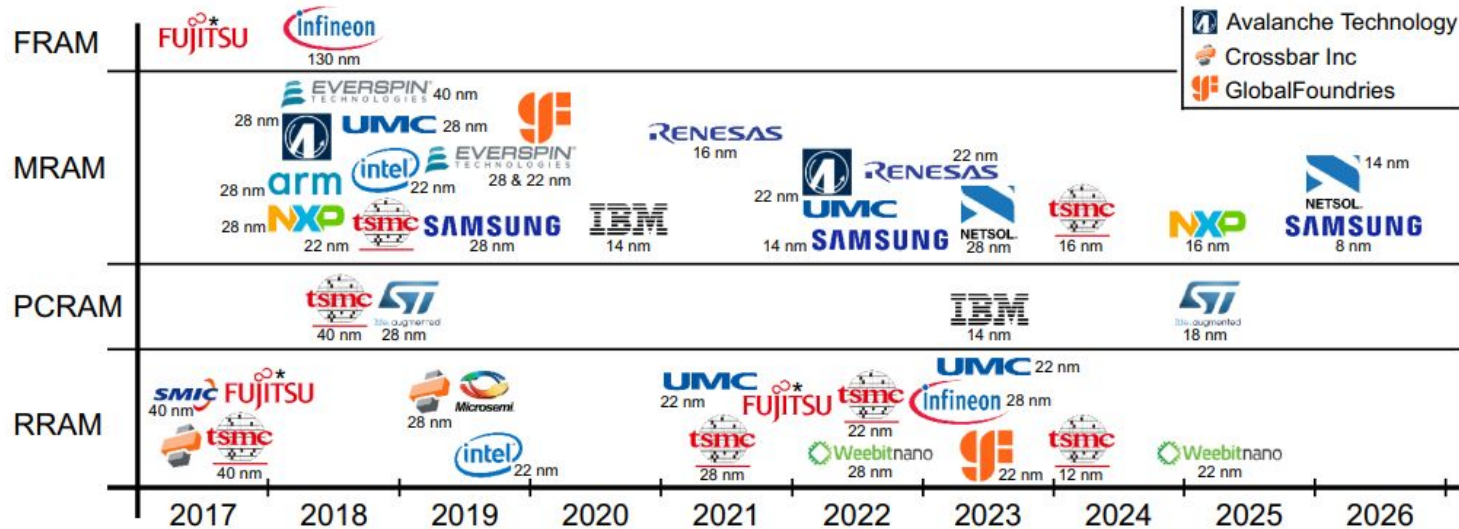
- **Replacement for eFlash, an embedded NVM (eNVM), which doesn't scale below 28nm.**
- Back-end-of-line (BEOL) integratable, allowing for higher density.
- Could even be used to replace volatile memory (VM) like SRAM and DRAM.

Hellenbrand, M., Teck, I. & MacManus-Driscoll, J.L. Progress of emerging non-volatile memory technologies in industry. *MRS Communications* **14**, 1099–1112 (2024). <https://doi.org/10.1557/s43579-024-00660-2>

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Industry Trends



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Industry Trends

TSMC - HfO_2 based in 40nm, 28nm, and 22nm. Demoed in 12nm FinFet in 2024.

Intel - Demonstrated 22nm FinFET RRAM with a TaO_x based oxide.

IBM - Looking into using both $\text{TaO}_x/\text{HfO}_2$ for analog-in-memory compute.

Hellenbrand, M., Teck, I. & MacManus-Driscoll, J.L. Progress of emerging non-volatile memory technologies in industry. *MRS Communications* **14**, 1099–1112 (2024). <https://doi.org/10.1557/s43579-024-00660-2>



Benchmarking Memory Technology

	SRAM	DRAM	Flash (NAND)	Flash (NOR)	RRAM
Volatile	Volatile	Volatile	Non-Volatile	Non-Volatile	Non-Volatile
Cell Size	~50 F ²	~6 F ²	~4 F ²	~10F ²	~6–12 F ²
Read Speed	Fast	Fast	Slow	Fast	Fast
Write Speed	Fast	Fast	Slow	Slow	Medium
Endurance	Unlimited	~10 ¹⁵	~10 ⁴ –10 ⁵	~10 ⁴ –10 ⁵	~10 ⁶ –10 ⁹
Write Voltage	V _{dd}	V _{dd}	~20V	~20V	V _{dd} + boost ~2V
Write Granularity	Per-bit	Per-bit	Block erase req.	Block erase req.	Per-bit
On Chip Usage	Yes	No	No	Yes	Yes

Competing RRAM Architectures: Operating Conditions

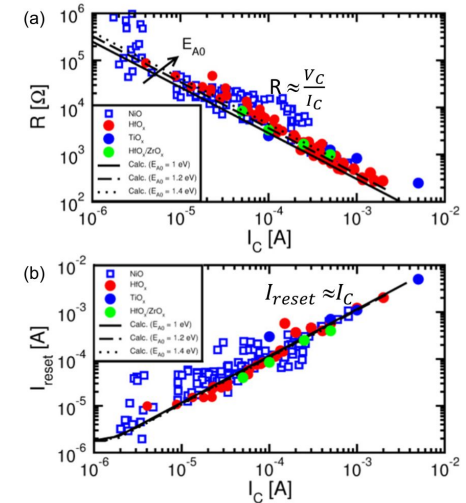


Operation	1T1R	Crossbar Array
READ	BL: 0.3-0.4 V, SL: 0 V, WL: VDD MOSFET: triode sense I_{cell} vs. I_{ref} HRS/LRS 10–1000× in HfO_2	WL = $-V_r/2$, BL = $+V_r/2$ I_{cell} vs I_{ref} Sneak: $I = (N-1) \cdot V_r / (2 \cdot R_{\text{LRS}})$ HRS/LRS must exceed array fan-out N for read margin
WRITE SET	BL: 2–2.5 V, SL: 0 V, WL: VDD MOSFET: saturation — used to limit sudden current jump $\sim R_{\text{LRS}} = V_c/I_{\text{cc}}$	WL = $+V_w/2$, BL = $-V_w/2$ Cannot limit sudden jump when channel conducts N-1 nodes on the WL and BL are half selected
WRITE RESET	SL: 1.5 V, BL: 0 V, WL: VDD MOSFET: saturation	BL = $+V_w/2$, WL = $-V_w/2$ N-1 nodes on the WL and BL are half selected

Niu et al., Sci. Rep. 6:28155 (2016) D. Ielmini, "Resistive switching memories based on metal oxides: mechanisms, reliability and scaling," Semiconductor Science and Technology, vol. 31, no. 6, p. 063002, 2016.

Why 1T1R?

RRAM Requirement	Transistor Design
Limit I_{cc} for SET (0-1) and Forming Voltage	1. Built in headroom to be in saturation during forming 2. Size so: $I_{ds} = I_{cc} = \frac{1}{2}\mu_n C_{ox}(W/L)(V_{gs} - V_t)^2$
Suppress Leakage Current from other Cells: $I_{sneak} = (N-1) \cdot V_r / (2 \cdot R_{LRS})$	1. Limit DIBL 2. Low SS = $(kT/q)\ln(10)(1 + C_d/C_{ox})$
Ensure Clean Read	1. Limit R_{on} of transistor $R_{LRS} \gg R_{on}$



D. Ielmini, "Resistive switching memories based on metal oxides: mechanisms, reliability and scaling," *Semiconductor Science and Technology*, vol. 31, no. 6, p. 063002, 2016.

Planar CMOS with RRAM

Sizing the Select Transistor

Series circuit constraint:

$$V_{bl} = V_{RRAM} + V_{ds}$$

Saturation:

$$V_{ds} > (V_{gs} - V_t) \text{ so } V_{bl} - V_{SET} > (V_{gs} - V_t)$$

W/L sizing from saturation model:

$$I_{ds} = \frac{1}{2} \mu_n C_{ox} (W/L) (V_{gs} - V_t)^2 = I_{cc}$$

LRS resistance:

$$R_{LRS} = V_c / I_{cc} \quad (V_c = 0.4 \text{ V for HfO}_2)$$

$$R_{on} = L / [\mu_n C_{ox} \cdot W \cdot (V_{gs} - V_t)] < R_{LRS}$$

Niu 2016

W = 1.14 μm , L = 0.24 μm

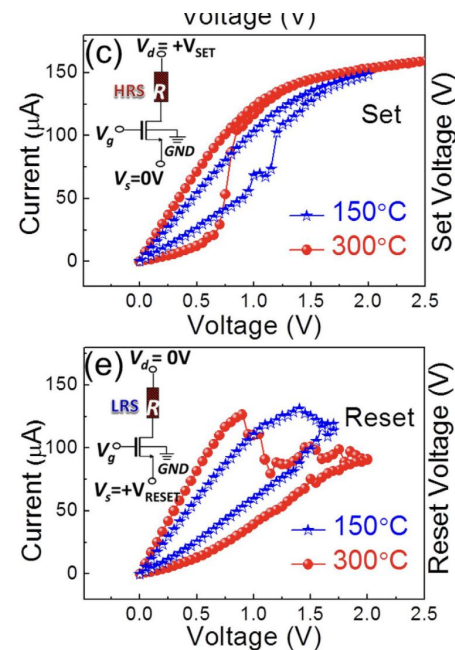
$V_g = 0.9 \text{ V}$ $I_{cc} = 10 \mu\text{A}$

$V_{SET} = 0.8\text{--}1.1 \text{ V}$, $V_{RESET} = 1.0\text{--}1.5 \text{ V}$

$R_{LRS} = 0.4 \text{ V} / 10 \mu\text{A} = 40 \text{ k}\Omega$

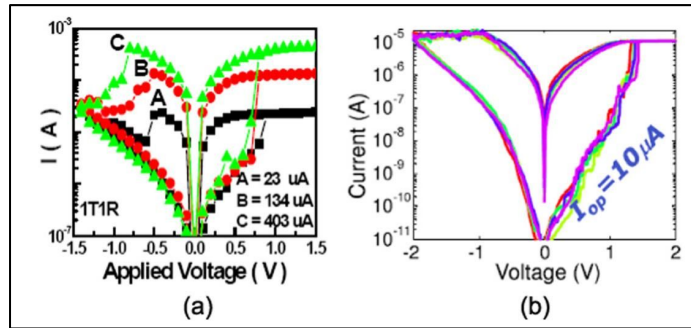
$R_{on} = 200 \Omega < 40 \text{ k}\Omega$

Forming: 3.2–4.0 V on BL



Niu, G., Kim, H.-D., Roelofs, R., Perez, E., Schubert, M. A., Zaumseil, P., Costina, I., & Wenger, C. (2016). Material insights of HfO₂-based integrated 1-transistor-1-resistor resistive random access memory devices processed by batch atomic layer deposition. *Scientific Reports*, 6, 28155.

Planar, FDSOI, FinFET Comparison RRAM



Parameter	Planar Bulk	FD-SOI	FinFET
VDD	1.0 V	0.9–1.0 V	0.8–0.9 V
I _{cc} Control	W/L + V _g	W/L + V _g + BG	N _{fg} + V _g
SS (mV/dec)	80–100	65–75	~65
Min. cell area	0.06 μm ²	0.066 μm ²	0.022 μm ²
Forming headroom	OK	Possible charge pump needed	Charge pump req
In Production eRRAM	TSMC 28nm	STMicro/GF	TSMC 12nm

D. Ielmini, Semicond. Sci. Technol. 31:063002 (2016), Fig. 5

Questions